exact size and shape of the extra dimensions—the approximate equations fall short. As we continue to sharpen our mathematical analysis and improve these approximate equations, determining the form of the extra dimensions is a prime—and in my opinion attainable—objective. So far, this goal remains beyond reach.

Nevertheless, we can still ask whether *any* choice of Calabi-Yau shape yields string vibrational patterns that closely approximate the known particles. And here the answer is quite gratifying.

Although we are far from having investigated every possibility, examples of Calabi-Yau shapes have been found that give rise to string vibrational patterns in rough agreement with Tables 12.1 and 12.2. For instance, in the mid-1980s Philip Candelas, Gary Horowitz, Andrew Strominger, and Edward Witten (the team of physicists who realized the relevance of Calabi-Yau shapes for string theory) discovered that each hole—the term is used in a precisely defined mathematical sense—contained within a Calabi-Yau shape gives rise to a *family* of lowest-energy string vibrational patterns. A Calabi-Yau shape with three holes would therefore provide an explanation for the repetitive structure of three families of elementary particles in Table 12.1. Indeed, a number of such three-holed Calabi-Yau shapes have been found. Moreover, among these preferred Calabi-Yau shapes are ones that also yield just the right number of messenger particles as well as just the right electric charges and nuclear force properties to match the particles in Tables 12.1 and 12.2.

This is an extremely encouraging result; by no means was it ensured. In merging general relativity and quantum mechanics, string theory might have achieved one goal only to find it impossible to come anywhere near the equally important goal of explaining the properties of the known matter and force particles. Researchers take heart in the theory's having blazed past that disappointing possibility. Going further and calculating the precise masses of the particles is significantly more challenging. As we discussed, the particles in Tables 12.1 and 12.2 have masses that deviate from the lowest-energy string vibrations—zero times the Planck mass—by less than one part in a million billion. Calculating such infinitesimal deviations requires a level of precision way beyond